

CSCE 5290 Natural Language Processing

Project Part1 Report- Group 11

Team Members:

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1.Work Load Summary:

Team member	Annotations done	Reviews division	Report part
Sampath	12	110	Coordinate among teammates and write the workload and worked on the challenging example 1 and also draft the combined report.
Abhiram	12	110	Worked the Challenging examples 2,5 finding why they are difficult and how it helps the model.
Govardhan	12	110	Worked on the part 3 cognition task difficulty part, i.e. identifying the task that model will find difficult to learn
Lavanya	12	110	Worked on part 4 identifying the new task by merging also worked on the example 2
Niipunn	12	110	Worked on the Identifying the new task and also worked on the example 3
Sushanth	12	110	Worked on the part 5 the most difficult tasks to combine.

We have done over around 75 Annotations, at first, we have done some Annotations where bounding labels are not labelled correctly, we rectified after the professor discussed in class and done some more so that we a total of around 50 annotations with all requirements.

The reviews we divided each person to do around 100-110 annotations per person we are working on . We have divided the workload equally and contributed equally, we participated in regular meetings and done report and while working on the reviews of annotations.

2. Selected Top Five Examples

Example 1:

Annotation 4 of 76

Question

is there any utensil to put sugar cubes in the pic?

Confidence in Question

1 3 5

Answer

yes

Confidence in Answer

1 3 5

Composition of

q_judge x q_type x

Add Your Answer

Confidence in Your Answer

1 3 5

Save Additional Answer

Annotated Image



Bounding box labels

tongs for the sugar cubes

Why this example is challenging:

The utensil (tongs) is tiny and is positioned at the back on the small and closed bowl that probably had sugar cubes, the question can also be regarded as commonsense, as well as judge and type reasoning. The model should be able to see the less prominent object taking into consideration the main food relations as there is milk item in the main bowl of the food. It also demands an insight into the role of the object and not merely identification.

How it helps training:

Models can be enhanced through this example to better detect small objects and use the environment learn how objects (sugar bowl and tongs) relate to their use.

Example 2:

Annotation 8 of 16

Question

What color is the leather sofa in the picture?

Confidence in Question

1 4 5

Answer

green

Confidence in Answer

1 3 5

Composition of

q_color x q_recognition x

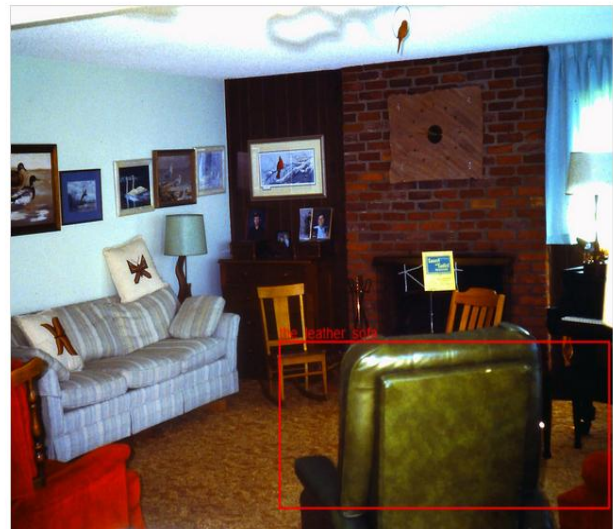
Add Your Answer

Confidence in Your Answer

1 3 5

Save Additional Answer

Annotated Image



Bounding box labels

the_leather_sofa

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Next

Why is this challenging:

The AI model should be able identify the leather sofa based on its **texture**(reflectiveness,glossy) , among multiple sofas in the room. Once the correct object has been identified, it must then be established in terms of its color and that involves integrating recognition and color understanding. The model identification of the texture cannot be easily identified without complex feature engineering programs.

How it helps training:

The example assists the models to learn fined-grained object recognition like the ability to differentiate between objects in terms of material or texture. It also enhances the capacity of the model to connect the property of the objects (leather) with attributes such as color resulting in enhanced visual reasoning.

Example 3:

Annotation 58 of 76

Question

Is there any snow or ice on any of the mountains shown ?

Confidence in Question

1 3 5

Answer

yes,yes

Confidence in Answer

1 4 5

Composition of

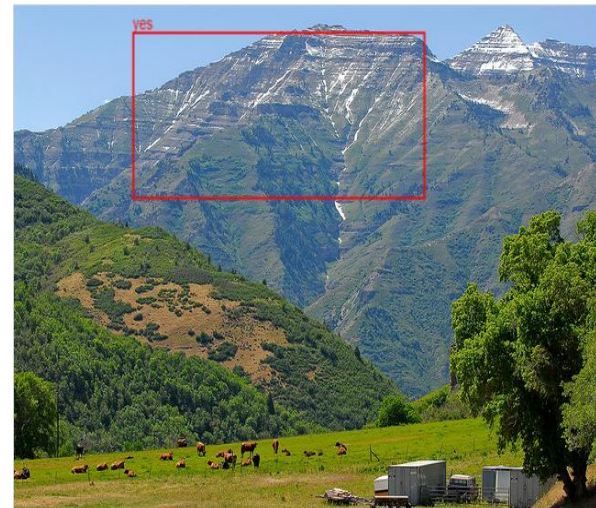
q_commonsense x q_recognition x q_color x

Add Your Answer

Confidence in Your Answer

1 3 5

Annotated Image



Bounding box labels

yes

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Why this example is challenging:

The model should identify small distant objects (snow) in big mountains, where objects are not so conspicuous. It must also discriminate between snow and such things of the same colour as light reflections or bare rock. The question also contains any and thus the model will have to examine the entire scene and not the one area.

How it helps training:

The example enhances fine-grained recognition of large scenes and trains the model to study the context of the entire image. It can also be used to learn to recognize the natural objects such as snow in different conditions which enhances the overall visual reasoning and generalization.

Example 4:

Annotation 41 of 76

Question

is this room messy or neatly organized?

Confidence in Question

1 5

Answer

messy

Confidence in Answer

1 5

Composition of

q_commonsense x q_judge x

Add Your Answer

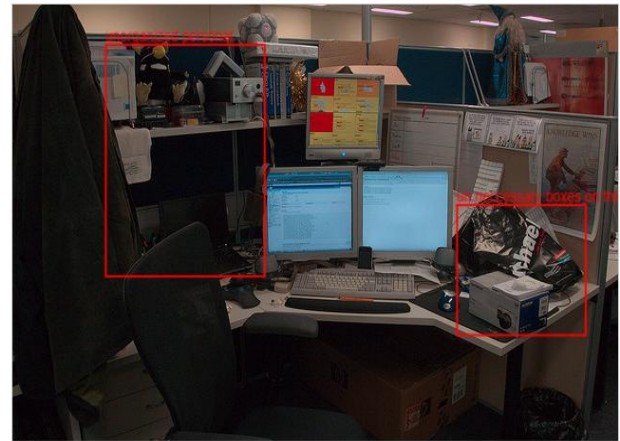
Confidence in Your Answer

1 5

3

Save Additional Answer

Annotated Image



Bounding box labels

un necessary boxes on the mouse pad which limits the movement of mouse
unorganized personal

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Direct Navigation

Why this example is challenging:

The model has to make a judgment regarding the scene subjectively rather than merely identify objects. It must examine various visual clues like cluttered items, unwanted boxes and haphazardly arranged personal belongings to deduce that there is a messy workplace. This involves the integration of visual knowledge and commonsense.

How it helps training:

This example assists models in enhancing contextual knowledge and reasoning on a scene level. It trains the model not to detect individual objects only but the overall organization of the objects and make sense of the state of an environment.

Example 5:

Annotation 27 of 76

Question

What is the white substance formed by the water as it crashes onto the shore?

Confidence in Question

1

5

Answer

Ocean foam

Confidence in Answer

1

5

Composition of

q_recognition x

q_action x

Add Your Answer

Confidence in Your Answer

1

5

Save Additional Answer

Annotated Image



Bounding box labels

This highlights the white foam created by the action of the waves breaking as they reach the shoreline

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Direct Navigation

3. Cognition Task Difficulty:

Based on our experience of the annotations in this project, we believe that commonsense reasoning is the most challenging task of cognition to teach a machine using a question-answer (QA) format. The commonsense questions assume that one needs to know about situations through general knowledge about the world that is not necessarily immediately visible in the image. While annotating, we have observed that it was more difficult to make explicit commonsense questions since the answers were usually subject to interpretation and not necessarily graphical evidence. Contrary to such kind of tasks as recognition, color, count or location, when the answer may be directly checked based on the picture, commonsense questions did not always have one single correct answer. Due to the fact that machines mostly learn visually based on the information available in the data, it is not easy to impart to them knowledge based on common human experience. Thus, to us in the context of making the annotation, commonsense reasoning was the hardest to capture in an effective format under QA.

4. Reflection on Cognition Task Definitions:

As a result of this project experience we learned that the so-called **common sense and causal** reasoning tasks can be combined in a single cognition task. In the process of annotation, the two tasks were not always easily differentiated since the two demand reasoning that goes beyond immediate observation in the image. Causal questions tend to require answers as to why something is occurring and commonsense questions tend to be based on the general knowledge of the world as to why a situation is occurring. Common sense knowledge is also needed in most instances to answer a causal question. Due to this overlap, similar question might occasionally be construed as falling under both categories leading to inconsistency when annotating them. To minimize this ambiguity, we can group these tasks into a new category of cognition, which we call the **Contextual Reasoning**. Questions that would be involved in this task would involve interpretation of the situation in the picture both through visual clues and general world knowledge. This would be a better way to combine them under this name and enhance uniformity in the dataset.

5. Challenging Task Compositions:

We found counting and commonsense reasoning to be one of the hardest to incorporate into a single question. Counting activities generally require the identification and counting of objects that are readily apparent in the picture and, as such it typically results in the provision of a straightforward numerical response. On the contrary, commonsense reasoning involves weaving out the situation with the aid of common sense which is not necessarily apparent in the picture. During the preparation of the annotations, we had trouble integrating these two tasks in a natural manner. Counting questions are usually answers that require a number, whereas commonsense questions demand the interpretation and reasoning of the scene. The nature of these two tasks necessitated the difference in thinking and thus, it was frequent to struggle to make both relevant on a question and at the same time to maintain the answer concise and accurate.